

PREFACE FOR FACULTY

Linear-phase FIR filters possess coefficient symmetry that allows hardware multipliers to be reduced by approximately 50%. In addition, the **Frequency-Sampling Realization** provides a recursive implementation of FIR filters using an all-zero comb filter and a bank of all-pole discrete resonators.

Pedagogical Strategy:

1. Prove how symmetry $h[n] = h[M - 1 - n]$ allows input samples to be pre-added before multiplication, cutting hardware multipliers from M to $\lfloor M/2 \rfloor + 1$.
2. Formulate the Linear-Phase Direct Form structure for both even and odd filter lengths.
3. Derive the Frequency-Sampling Transfer Function:

$$H(z) = \frac{1 - z^{-M}}{M} \sum_{k=0}^{M-1} \frac{H[k]}{1 - W_M^{-k} z^{-1}}$$

4. Explain the physical mechanism: The comb filter $(1 - z^{-M})$ places M zeros on the unit circle at $z = e^{j2\pi k/M}$, while the resonator poles cancel the zeros at specific bin locations.
5. Address practical engineering issues: Pole-zero cancellation sensitivity on $|z| = 1$ and stability under coefficient quantization (damping factor $r = 0.999 \approx 1$).

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Design** Linear-Phase FIR structures with 50% reduced multiplier count.
2. **Derive** the Frequency-Sampling filter realization equations from the IDFT formulation.
3. **Construct** signal flow graphs for frequency-sampling FIR filters.
4. **Evaluate** computational efficiency for narrowband filters with few non-zero DFT bins.

2. MATHEMATICAL FOUNDATIONS

2.1 Linear-Phase Multiplier Reduction

For a symmetric FIR filter of length M ($h[n] = h[M - 1 - n]$):

- **Odd Length M (Order $N = M - 1$ Even):**

$$y[n] = \sum_{k=0}^{(M-3)/2} h[k](x[n-k] + x[n-M+1+k]) + h\left[\frac{M-1}{2}\right]x\left[n - \frac{M-1}{2}\right]$$

Total multipliers: $\frac{M+1}{2}$.

- **Even Length M (Order $N = M - 1$ Odd):**

$$y[n] = \sum_{k=0}^{M/2-1} h[k](x[n-k] + x[n-M+1+k])$$

Total multipliers: $\frac{M}{2}$.

2.2 Frequency-Sampling Structure Derivation

Express the FIR transfer function in terms of its M -point DFT samples $H[k]$:

$$h[n] = \frac{1}{M} \sum_{k=0}^{M-1} H[k] W_M^{-nk}$$

$$H(z) = \sum_{n=0}^{M-1} h[n] z^{-n} = \sum_{n=0}^{M-1} \left(\frac{1}{M} \sum_{k=0}^{M-1} H[k] W_M^{-nk} \right) z^{-n} = \frac{1}{M} \sum_{k=0}^{M-1} H[k] \sum_{n=0}^{M-1} (W_M^{-k} z^{-1})^n$$

Using the finite geometric sum $\sum_{n=0}^{M-1} u^n = \frac{1-u^M}{1-u}$, and since $(W_M^{-k} z^{-1})^M = W_M^{-kM} z^{-M} = 1 \cdot z^{-M} = z^{-M}$:

$$H(z) = \frac{1 - z^{-M}}{M} \sum_{k=0}^{M-1} \frac{H[k]}{1 - W_M^{-k} z^{-1}}$$

- **Comb Filter:** $H_{\text{comb}}(z) = \frac{1-z^{-M}}{M}$ (FIR, zeros at $z = e^{j2\pi k/M}$).
- **Resonator Bank:** $H_k(z) = \frac{H[k]}{1 - W_M^{-k} z^{-1}}$ (All-pole, pole at $z = e^{j2\pi k/M}$).

3. WORKED NUMERICAL EXAMPLES

Example 16.1: Linear-Phase FIR Structure

Problem: Draw the linear-phase direct form structure for an FIR filter with impulse response

$$h[n] = \underset{\uparrow}{1}, 2, 3, 2, 1 \quad (M = 5).$$

Solution:

- Length $M = 5$ (Odd).
- Symmetry: $h[0] = h[4] = 1$; $h[1] = h[3] = 2$; $h[2] = 3$.
- Output equation:

$$y[n] = 1 \cdot (x[n] + x[n-4]) + 2 \cdot (x[n-1] + x[n-3]) + 3 \cdot x[n-2]$$

- **Hardware Count:**
 - Multipliers: $\frac{5+1}{2} = 3$ (compared to 5 in standard direct form).
 - Adders: 2 (pre-additions) + 2 (accumulation) = 4.
 - Delays: 4 registers z^{-1} .

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Derive the Frequency-Sampling structure for an FIR filter. Explain how pole-zero cancellation is achieved. (8 Marks) (b) Draw the linear-phase realization for an FIR filter of length $M = 6$ with $h[n] = \underset{\uparrow}{-1}, 2, 4, 4, 2, -1$. (7 Marks)

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
 - Complete IDFT substitution and geometric series derivation (5 Marks)
 - Explanation of comb zeros cancelling resonator poles on $|z| = 1$, and damping factor $r \lesssim 1$ for numerical stability (3 Marks)
- **Part (b):**
 - $M = 6$ (Even length, symmetric):
 $y[n] = -1(x[n] + x[n - 5]) + 2(x[n - 1] + x[n - 4]) + 4(x[n - 2] + x[n - 3])$ (3 Marks)
 - Multipliers required: $6/2 = 3$.
 - Neatly drawn SFG with 5 delay elements, 3 pre-adders, 3 multipliers, and accumulator bus (4 Marks)

5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

# Verify linear-phase convolution equivalence
h = np.array([1, 2, 3, 2, 1])
x = np.array([1, 1, 1, 1, 1])
y = np.convolve(x, h)
print("Convolution Output:", y)
```